

MIDDLE ROCK LIGHT, AK, USA

WMO: 994670

Lat: 61.0811N

Lon: 146.6530W

Elev: 0

StdP: 101.33

Time Zone: -9.00 (NAK)

Period: 02-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -11.3	(c) -9.6	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) 17.9	(k) -5.5	(l) 16.0	(m) -5.4	(n) 9.2	(o) 50	(p) 0.778

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 4.2	(c) 17.5	(d) N/A	(e) 16.1	(f) N/A	(g) 15.1	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) 6.8	(p) 200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) N/A	(b) N/A	(c) N/A	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 15.1	(b) 12.1	(c) 10.2	(d) DB	(e) -12.9	(f) 21.6	(g) 1.8	(h) 3.4	(i) -14.2	(j) 24.1	(k) -15.2	(l) 26.0	(m) -16.2	(n) 27.9	(o) -17.5	(p) 30.4	(q) N/A
(a) 15.1	(b) 12.1	(c) 10.2	(d) WB	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A	(q) N/A

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	4.6	-2.2	-1.4	-0.4	3.4	7.9	10.7	12.0	11.4	8.9	5.2	0.2	-0.9		
	DBStd	5.86	4.05	3.15	3.06	2.46	2.36	1.91	1.69	1.80	1.67	2.12	3.61	3.15		
	HDD10.0	2128	377	319	321	197	74	12	3	4	41	149	293	338		
	HDD18.3	5009	635	552	579	447	324	229	196	214	284	408	543	597		
	CDD10.0	161	0	0	0	1	8	33	64	48	7	0	0	0		
	CDD18.3	1	0	0	0	0	0	1	0	0	0	0	0	0		
	CDH23.3	3	0	0	0	0	0	2	1	1	0	0	0	0		
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0		
(14) Wind	WSAvg	4.7	6.0	5.4	5.7	3.8	4.2	4.3	4.2	4.0	3.7	4.0	5.8	5.5		
(15) Precipitation	PrecAvg	1125	101	94	68	57	51	57	84	118	153	126	99	113		
	PrecMax	1401	263	235	148	111	112	100	132	281	260	320	239	254		
	PrecMin	771	11	0	15	13	12	18	42	16	69	55	2	25		
	PrecStd	163	58	53	43	27	27	21	26	64	53	68	65	57		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.9	4.7	6.8	12.6	16.8	19.7	20.0	20.7	15.2	11.1	7.7	5.2		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	DB	4.3	3.6	5.3	9.6	14.5	16.5	17.0	17.3	13.5	9.7	6.0	4.2		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	DB	3.2	2.9	4.3	7.8	12.8	14.7	15.9	15.8	12.4	8.7	5.1	3.4		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	10%	DB	2.4	2.2	3.4	6.5	11.3	13.6	15.1	14.6	11.4	7.9	4.3	2.7		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(35) Mean Daily Temperature Range	MDBR		2.9	2.8	3.2	3.5	3.7	3.6	4.2	4.4	3.7	2.9	2.8	2.8		
	5% DB	MCDBR	3.6	3.4	4.0	5.3	6.3	6.0	6.1	6.6	5.4	4.2	3.1	3.5		
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(40) Clear-Sky Solar Irradiance	taub		0.267	0.328	0.333	0.388	0.411	0.401	0.410	0.405	0.348	0.326	0.282	0.263		
	taud		2.002	2.012	2.049	1.951	2.019	2.182	2.214	2.218	2.338	2.179	1.983	1.901		
	Ebn at Noon		421	604	761	785	802	823	804	771	759	614	378	236		
	Edh at Noon		44	83	111	149	151	131	124	114	84	69	43	30		
(44) All-Sky Solar Radiation	RadAvg		0.28	0.97	2.52	4.30	5.42	5.65	5.04	3.84	2.47	1.13	0.40	0.15		
	RadStd		0.03	0.14	0.26	0.35	0.41	0.32	0.32	0.37	0.33	0.12	0.07	0.02		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46) Station Only		N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A
(47) Regional (22 neighbors)		N/A	+1.60	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page